MICHIGAN MEDICINE
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Vancomycin Dosing and Monitoring in Patients on Pediatric Services

Patient Population/Unit/Service	Dosing Recommendations
Dosing recommendations for patients with normal renal function (CrCl ≥ 60 mL/min) except in the following situations: • **Congenital heart center (CHC) patients** • **Cystic fibrosis (CF) patients** • **Extracorporeal membrane oxygenation (ECMO) patients** • **Low muscle mass** • **Neonatal intensive care unit (NICU) patients**	If patient previously received vancomycin, review the previous dosing regimen, levels, indication, and patient information (including age and renal function). Consider using previous regimen if clinically appropriate. Infants > 5000 g, children, and adolescents: 20 mg/kg/dose IV q8h (max initial dose: 1250 mg) • Consider rounding vancomycin doses to the nearest premade size when clinically appropriate (within 10% of ordered dose) • Young adults (18 years and older) admitted to a pediatric service can be dosed using this guideline • For infants ≤ 5000 g and not on CHC service, refer to NICU dosing; refer to CHC dosing for patients on a cardiology service
Patients with renal insufficiency For infants ≤ 5000 g not on CHC service, refer to NICU dosing	CrCl 40 to < 60 mL/min: < 12 years: 15 mg/kg IV q8h ≥ 12 years: 10 mg/kg IV q8h CrCl 30 to < 40 mL/min: < 12 years: 15 mg/kg IV q12h ≥ 12 years: 10 mg/kg IV q12h CrCl < 30 mL/min: < 12 years: 15 mg/kg IV q24h* ≥ 12 years: 10 mg/kg IV q24h* *Check random level between 12-24 hours after first dose. Redose for level < 15 mcg/mL
Dosing recommendations for children on CRRT, peritoneal dialysis, or hemodialysis For patients on NICU service line, refer to NICU dosing	CRRT: Dose per the following equation: $Effluent\ rate\ \left(\frac{mL}{m^2*hr}\right) = \frac{Dialysate\ rate\ \left(\frac{mL}{hr}\right) + Replacement\ rate\ \left(\frac{mL}{hr}\right)}{BSA\ (m^2)}$ • < 2000 mL/m^2/hr: 10 mg/kg/dose IV q12h (max 1000 mg/dose) • ≥ 2000 to <4000 mL/m^2/hr: 15 mg/kg/dose IV q8h (max 1000 mg/dose) • ≥ 4000 mL/m^2/hr: 20 mg/kg/dose IV q8h (max 1000 mg/dose) -Please consider native UOP and potential for additional clearance -Obtain levels prior to 2nd or 3rd dose depending on clinical scenario Peritoneal dialysis: 10 mg/kg/dose IV q24h Hemodialysis: 10 mg/kg/dose IV x1 post dialysis
Low muscle mass (e.g., muscular dystrophy, cerebral palsy, spinal muscular atrophy)	15 mg/kg/dose q8h

CHC patients Including those on ECMO If patient meets multiple criteria for vancomycin dose adjustment (open chest prophylaxis, within 72 hours of cardiac surgery, depressed cardiac function, ECMO, and/or renal insufficiency), select the lowest of those dosing regimens. Empiric dosing for dialysis patients (CRRT, iHD, PD) should follow the dosing recommendations in that section.	Open chest prophylaxis or patients within 72 hours of cardiac surgery: 10 mg/kg/dose IV q12h Patients with depressed cardiac function/heart failure* AND ≥ 72 hours since last cardiac surgery: 10 mg/kg/dose IV q8h Patients with normal cardiac function AND ≥ 72 hours since last cardiac surgery: 15 mg/kg/dose IV q8h If CrCl < 60 mL/min, please refer to dosing recommendations for Patients with renal insufficiency *Depressed cardiac function/heart failure indicated by patient being on inotropes (e.g., milrinone) OR at least two of the following: ACE-inhibitor, beta-blocker, digoxin, spironolactone
ECMO (PICU) For infants ≤ 5000 g not on CHC service, refer to NICU dosing	Children with normal renal function (CrCl ≥60 mL/min): 15 mg/kg/dose IV q8h (max initial dose: 1250 mg) Children with renal insufficiency (CrCl <60 mL/min): Refer to Patients with renal insufficiency section above
Cystic fibrosis	Children < 12 years: 20 mg/kg/dose IV q6h (max initial dose: 900 mg) Children ≥ 12 years: 20 mg/kg/dose IV q8h (max initial dose: 1250 mg)
NICU	Less than 1200 g AND 14 days or younger: 15 mg/kg/dose IV q18h Less than 1200 g AND older than 14 days: 15 mg/kg/dose IV q12h 1200-2000 g AND 14 days or younger: 15 mg/kg/dose IV q12h 1200-2000 g AND older than 14 days: 15 mg/kg/dose IV q8h 2000-5000 g AND 7 days or younger: 15 mg/kg/dose IV q12h 2000-5000 g AND older than 7 days: 15 mg/kg/dose IV q8h More than 5000 g: 20 mg/kg/dose IV q8h ECMO: 15 mg/kg/dose IV q18h Therapeutic hypothermia (cooling): 15 mg/kg/dose IV q18h Peritoneal dialysis: 10 mg/kg/dose IV q24h

Goals of Therapy for Vancomycin:

	Therapeutic Goals
Vancomycin	● AUC is the preferred method of vancomycin monitoring ● Goal AUC is 350-530 regardless of MIC and should not be adjusted for MICs less than or equal to 1 ● Open chest prophylaxis: ○ Trough 5-10 mcg/mL ● Redose if pre-dialysis: ○ < 20 mcg/mL ● Dosing by levels: ○ Trough 10-15 mcg/mL

Initiating Vancomycin Therapy:

- Doses should be based on DOSING weight.
- Avoid initial doses > 3750 mg per DAY.

Monitoring within 48 hours of starting vancomycin:

- Vancomycin levels should be unnecessary if therapy is not anticipated to exceed 48 hours.
- Do not check vancomycin concentrations within the first 48 hours except in the following situations:

Clinical Situation	**Monitoring Recommendation**
Approximately 90% of patients will have vancomycin discontinued within 48-72 hours and most patients do not require levels	
Documented Gram-positive infection Septic shock Weight > 100 kg Children with low muscle mass (e.g., muscular dystrophy, cerebral palsy, spinal muscular atrophy)	• Obtain 2 vancomycin levels at steady state and calculate AUC to achieve goal AUC of 350-530 • Obtain a peak level ~2 hours after the END of infusion and a trough prior to the next dose for most patients to calculate AUC
Significant acute changes in renal function, CrCl < 30 mL/min, therapeutic hypothermia, ECMO, AKI, or neonates < 72 hours old whose mothers received peri-partum vancomycin	• Obtain a vancomycin level and dose per level • Monitor random levels in patients and re-dose when level < 15 mcg/mL
New start ECMO Peritoneal dialysis Cooling protocol Concern for renal dysfunction as indicated by the following: • Low urine output (< 1 mL/kg/hr) • SCr > 130% of baseline • Escalating vasopressor requirements	• Consider obtaining a vancomycin trough prior to second dose (aka “safety level”)

Monitoring after 48 hours of starting vancomycin:

Use the following table to guide monitoring of vancomycin based on the patient's clinical status:

Clinical Situation	Monitoring Recommendation
NICU	• Trough only monitoring may be preferred for patients < 1000 g OR in first week of life due to low blood volume and/or potentially rapidly fluctuating renal function • However, in patients with stable renal function, AUC based monitoring is recommended for invasive MRSA infections to optimize treatment efficacy
Patients with stable renal function (including patients with CKD and receiving CRRT)	• Obtain 2 vancomycin levels at steady state and calculate AUC to achieve goal AUC of 350-530 • Obtain a peak level ~2 hours after the END of infusion and a trough prior to the next dose for most patients to calculate AUC • Document individualized trough range that corresponds to AUC of 350-530 for that patient
Patients on conventional hemodialysis	• Check pre-HD level • Redose if pre-HD level < 20 mcg/mL
CHC patients within 72 hours of surgery	• Check trough concentration • Redose for trough < 10 mcg/mL
Patients who have fluctuating fluid and/or renal status	• Use clinical judgement to determine monitoring strategy • It is reasonable to perform AUC or trough-based monitoring. The instability of renal clearance or volume of distribution should be taken into account when evaluating levels and subsequent dosing

Refer to the following table for recommendations on frequency of ordering vancomycin levels and serum creatinine:

Clinical Situation	Monitoring Recommendation
Subsequent levels should be drawn every 1-7 days, and serum creatinine should be monitored at least every 48 hours during entire course of vancomycin therapy. Avoid evening and overnight levels if clinically stable.	
Patients with changing fluid status or renal function	• Obtain levels every 1-3 days • Monitor 2 vancomycin levels to facilitate AUC calculation, when possible • In patients receiving one-time doses (i.e., dosing by level), monitor random levels and re-dose when level < 15 mcg/mL
Patients with stable fluid status and renal function requiring long-term therapy	• Obtain levels every 5-7 days, after initial level(s) are therapeutic • Once a patient is on a stable dose with an AUC between 350 and 530, monitoring of vancomycin troughs may be acceptable in patients with stable fluid status and renal function

NOTES

- Dose should not exceed 100 mg/kg/day at any point in therapy.
- Vancomycin doses may not need to be increased for a patient with clinical improvement and an AUC below 350. There is likely no benefit to increasing doses to target the higher end of the 350 – 530 range, and dose increases within this range may increase the risk of toxicity. Alternative therapies should be considered for patients failing to respond to vancomycin. Clinical judgement must be applied.
- ID consult should be ordered for all patients with *S. aureus* bacteremia or patients with *S. aureus* infection who do not improve on vancomycin.
- Vancomycin infusion reaction, characterized by flushing, redness of the trunk, and itching, may occur during or shortly after the infusion. Management should include prolonging the infusion time to 2-4 hours. Could also consider diphenhydramine.

Michigan Medicine AUC Calculator:
https://drive.google.com/uc?export=download&id=1DH4tWDS1PX5KPKtMH5J9XdG3vbPXyOaE

Additional approvals:

CW Decentralized Pharmacist Committee: 05/2026

Last revised: 04/2026	**Antimicrobial Stewardship Subcommittee:** 04/2026	**P&T:**

References	
	Citation

Revision History	
Date	**Revision**
04/2021	Adjusted NICU dosing, added low muscle mass dosing
05/2021	Adjusted NICU, CF, and general population dosing
09/2022	Adjusted CHC dosing, added AUC vs. trough monitoring guidance for NICU, adjusted pre-HD level for redosing, added CRRT dosing, clarified timing of peak levels, changed language regarding vancomycin infusion reaction
04/2026	Adjusted AUC goals, added guidance for patients ≥ 18 years old covered by a pediatric service line, clarified how to select empiric dose for CHC patients

The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experiencing a medical emergency, call 911 immediately. These guidelines should not replace a provider's professional medical advice based on clinical judgment, or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time to time change. The authors of these guidelines have made all attempts to ensure the accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through an independent source.